

Visualizations for Subspace Clustering

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Abstract—Analyzing high dimensional data is a challenge. Subspace clustering is one of the approaches to comprehend such data. Subspace clustering-based analysis methods have gained popularity as they reveal patterns in all combinations of dimensions. Visual analytics plays a critical role to represent knowledge gained from extracted patterns. It enables a user to analyze, interpret and validate data visually. This paper presents a survey on visualization techniques that use subspace clustering patterns to understand high dimensional data. As many Visualizations exist to understand and interact with subspace clustering results, the main goal of this paper is to (I) Categorize these visualization techniques into different modules based on their underlying functionality. This provides an overview of working pipeline to understand high dimensional data through visualizations, (II) Enumerate and discuss existing methods in each category and state their pros and cons, (III) Summarize all the methods at one place showing current state of art and deducing the best-suited visualizations to comprehend different aspects of high dimensional data. This survey can assist researchers to choose the right method for a given problem and design new advanced visualization to get better insight into data. Hence giving research a direction for future.

Index Terms—High Dimensional Data, Subspace Clustering, Information Visualization, Visual Analytics.



1 INTRODUCTION

TECHNOLOGICAL development has led to the generation of a vast amount of high dimensional data. Therefore, datasets are growing both in terms of size (i.e. the number of records) and complexity (i.e. the number of dimensions). It is paramount to analyze high dimensional data. Raw data is not of much use, and it has to be processed in order to extract knowledge or meaningful information. To comprehend high dimensional data many approaches have been proposed in the past decade such as *Feature selection* [1], *Dimensionality reduction* [2], *Clustering* [3] etc. (as illustrated in Fig.1). Although feature selection plays a vital role in removing redundant and irrelevant features [4] but patterns that may help in comprehending high dimensional data may exist in any combination of dimensions. Therefore, it is not very popular approach. Similarly, dimensionality reduction techniques like PCA [5], t-SNE [6], MDS [7], SOM [8] etc., although it maps data into lower dimension while preserving the structure of data but tracing the meaning of pattern in original space is again a challenging task. Therefore, among these, clustering is one of the most dominant approaches to understand high dimensional data as it identifies natural groupings in the data. In addition, the relationships between data objects can be accessed by clustering algorithms, as similar data objects are grouped together and dissimilar once are separated.

Experimentally it has been proved that clustering in full-space (i.e. considering all dimensions simultaneously), is not effective for the high dimensional dataset [9]. The reason be-

hind this is the is *curse of dimensionality* [10] [11] i.e. the available data becomes sparse very quickly as the dimensionality of the dataset increases. This sparsity makes it difficult to cluster data. Therefore, for a dataset with large feature space, interesting patterns exist only in the subset of dimensions i.e subspaces. This survey focus on subspace clustering [3] [12] [13] based analysis methods to comprehend high dimensional data. The central idea of Subspace clustering is that clusters may exist in all combinations of attributes. Hence, subspace clustering associates each cluster with its (1) *Cluster members* i.e. subset of points forming cluster, (2) *Subspace* i.e. subset of dimensions in which cluster is formed. Once the meaningful groups of points sharing common characteristics in subspaces are identified by subspace clustering. The extracted patterns then need to be presented in such a way that it assists human understanding. There are several methods in pursuit of knowledge representation including data visualization, auditory representations textual summaries [14]. This work focus on *data visualization* aspect of knowledge representation. To extract knowledge from patterns obtained, meaningful visualization is very crucial. Visual analytics enables a user to analyze, interpret and validate data visually. Also interactive data visualization [15] has the potential to foster interactions [16], which bridges the gap between automated results and cognitive abilities of human users [17].

For all these reasons we believe visualizations using subspace clustering are valuable for exploration of high dimensional data. Hence is worth studying. This paper presents a survey on visualization techniques that use subspace clustering patterns to understand high dimensional data. A lot of approaches exists to visualize, explore and interact subspace clusters but no categorization is available to show a global picture of this area. This work attempts to categorize, compare, discuss and state pros and cons of subspace clustering based visualizations to understand different aspects of high dimensional data. It contributes a comprehensive study that can bring this research field

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forward.

This paper is organized as follows: Section(II) presents an overview of working pipeline to understand high dimensional data through visualizations on subspace clusters. For this, it categorizes these visualization techniques into different modules based on their underlying functionality. (II) elaborates on subspace analysis methods required to reduce exponential large number of subspaces to facilitate knowledge discovery, Section(III) explains subspace cluster analysis methods which are required as for high dimensional data meaningful information resides in sub dimensional space, Section(IV) highlights advanced Visualizations on high dimensional data, Section(V) concentrates on accessing subspace cluster quality, and Section(VI) concludes the paper with future vision and two tables, one of them summarizes different techniques discussed in this paper and other reveals visualizations for comprehending distinctive aspects of high dimensional data.

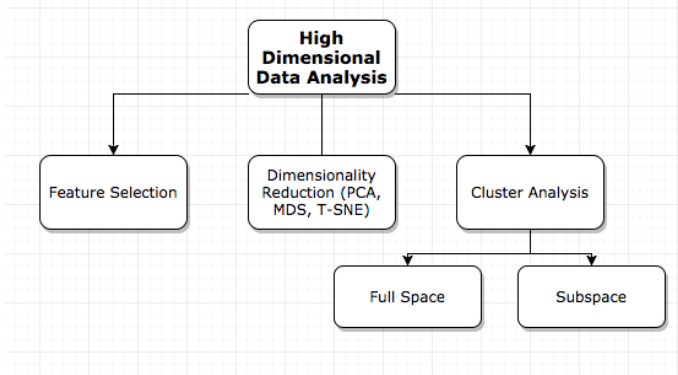


Fig. 1. Different approaches for High Dimensional Data Analysis

2 SURVEY METHOD AND OVERVIEW

Analyzing high dimensional data is an open research problem and has attracted numerous research efforts. We conducted a comprehensive study of relevant work in the emerging field of high dimensional data visualizations based on subspace clustering methods dedicated to explore navigate and interpret high dimensional data. This survey focuses on the works from the period between 2004 to 2017. To ensure this survey covers all pioneering works in this area, we started with an initial set of papers and searched through references and citations. Due to space limitations, we then performed regressive selection process to shortlist subset of papers that have a significant impact in this research field. Different approached in the shortlisted subset of papers are categorized, compared and discussed. This work is presented to support in-depth and overview study of understanding high dimensional data using data visualization on subspace clusters.

Fig.2) provides a global picture of working pipeline of visualizations on subspace clusters. The input to this Pipeline is high dimensional data which need to be explored and understood. The intensive process of categorizing visualizations is done in two phases. The first phase focuses on pre-processing input data to discover interesting subspaces having clusters. There are 2^d possible subspaces for d dimensional data. Therefore considering all possible subspaces

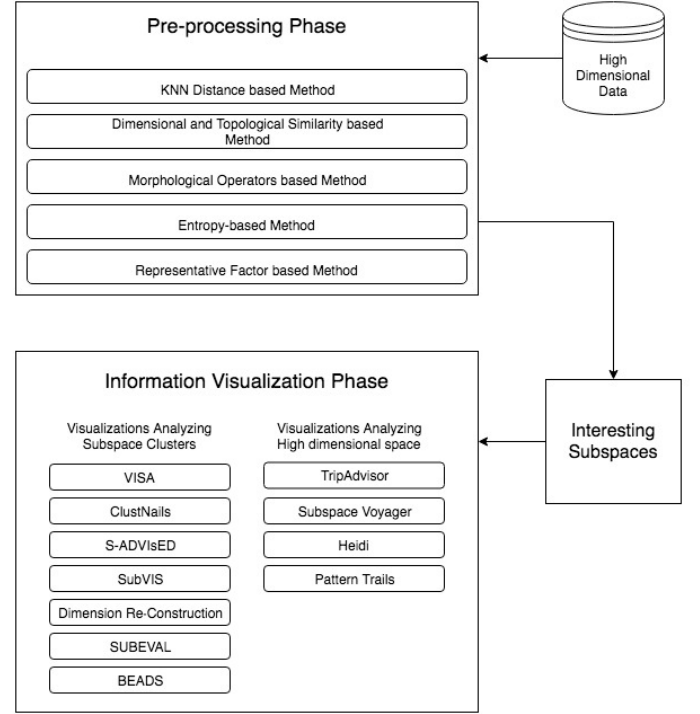


Fig. 2. Working pipeline for high dimensional data visualizations on subspace clusters

directly for analysis and visualization can lead to cluttered and uninterpretable results. Hence, preprocessing phase of this pipeline has a remarkable impact on further explorations. Different methodologies in this phase are discussed in section 3. Numerous visualizations have been proposed on interesting subspaces. The next phase is committed to the categorization of these visualizations based on the functionality they perform or domain of information they explore. There are two broad categories to which visualizations are mapped:

- Visualizations analyzing and comparing subspace cluster properties.
- Visualizations analyzing high dimensional space.

This paper gives an insight into different methods in each of the two categories by stating their core idea, advantages and disadvantages. It also attempts to compare different techniques to facilitate researchers to choose best-suited visualizations according to their requirement.

3 PRE-PROCESSING PHASE: FILTERING SUBSPACES OF INTEREST

In high Dimensional data, the number of possible subspaces which might contain relevant information of interest increases exponentially with dimensions. Considering all possible subspaces of high dimensional data makes the analytics task computationally expensive. Moreover, when the subspace clustering algorithms are applied to high dimensional data, they generally report a substantial number of clusters, which may contain significant redundancy or overlapping in terms of records contained and/or dimensions involved in subspaces associated with respective clusters.

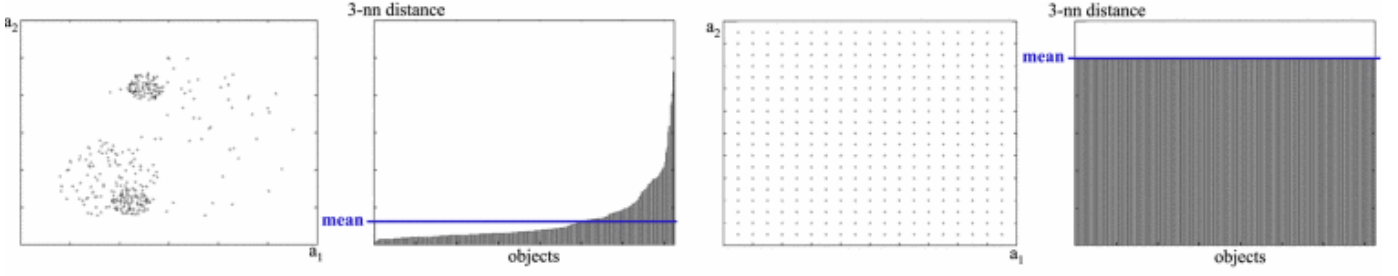


Fig. 3. KNN-distance of two datasets are plotted. Datasets with variable KNN-distance of objects (on left) exhibits better hierarchical clustering structures as compared to the one with with stable KNN-distance (on right) (This image is taken from [18])

This redundancy in results needs to be removed in order to get meaningful and understandable results. Therefore, most of the visualizations based on subspace clustering filters subspaces of interest as a preprocessing step. To cope with the above-stated problem, a lot of research has been done in recent years which aim at reducing a large number of subspaces to a manageable number of subspaces on which further analysis can be performed. However, some of the contributions [19] [20] in this area are not desirable as they are highly sensitive to input parameters (that are hard to determine) and/or are based on global density thresholds. This section of the paper focuses on the class of Subspace Analysis algorithms that deal with the problem of identifying interesting subspaces and clusters in high dimensional data.

3.1 K-Nearest Neighbour Distance based Method

Surfing [18] algorithm is used for the early abandoning of uninteresting subspaces. This is achieved by finding all subspaces that are interesting for clustering and ranking them by their relevance. The ranking is done based on quality criteria based on k -nearest neighbour distance (k -nn distance) of objects. This algorithm revolves around calculating interestingness of subspace which depends on hierarchical clustering [21] structure exhibited by data in that subspace. First k -nn distance of each object " o " is calculated, $nn_Dist_k^S(o)$ (which is distance between object o and k^{th} nearest neighbour of " o ") indicates how densely data is distributed around object in subspace S . Smaller this distance, more densely the data objects are distributed around the object under consideration. For each subspace, $nn_Dist_k^S(o)$ distances of objects are then plotted as shown in Fig.3. Interesting subspaces are the ones which show high variability in these distances. Because in such subspaces hierarchical clustering structure will be considerably clearer in contrast to subspaces where these distance are almost uniform. This is well demonstrated in Fig.3. Quality of each subspace is calculated based on the mathematical expression given below:

$$quality = \frac{1}{2} \sum_{\forall o} mean_S - nn_Dist_k^S(o)$$

The advantage of this algorithm is that it is almost parameter-less i.e user is not required to specify parameters which are hard to predict such as number of clusters, density thresholds etc. Moreover, it does not rely on global density threshold. It does not assume any specific clustering

structure and is dimensionality unbiased. Thus addresses the unsupervised notion of clustering in the best possible way. The shortcoming of this algorithm is its running time. Time complexity to find k -nn distance for each object in the dataset is $O(N^2)$ where N is the number of objects in the dataset. If m subspaces are analyzed by SURFING algorithm then total time complexity is $O(m.N^2)$ and worst case time complexity is $O(2^d.N^2)$ but in practice, very few percentage of total subspaces are examined by this algorithm. Therefore, it is a promising subspace search algorithm and is used as a prerequisite step for constructing several visualizations on high dimensional data.

3.2 Dimensional and Topological Similarity based Method

It is an interestingness guided subspace search algorithm based on subspace similarity function to construct candidate set of subspaces [22]. It takes high dimensional data as input, applies subspace search algorithm like SURFING which gives subspaces interesting for clustering. Although it reduces the number of subspaces still there may be a lot of redundant subspaces. This redundancy is tackled by leveraging dimensional and topological similarity in subspaces. Therefore the next step is to apply similarity-based grouping on subspaces. To accomplish this hierarchical agglomerative grouping is applied to yield a smaller set of interesting and mutually different groups. Grouping stands on:

- Similarity based on overlapping dimensions.
- Similarity based on the topological representation of data in subspaces.

For each group, subspace with the lowest dimensionality and highest interestingness score is filtered as representative. Further, the interactive exploration is proposed over group representatives of subspaces. For interactive exploration, subspaces are represented as scatter plots that allow rich topological comparisons. Scatter plots are generated using projection techniques like PCA, MDS, t-SNE etc. Interactive support allows users to compare and relate subspaces based on constituent dimensions and cluster objects involved.

This system allows exploration of high dimensional data and introduces a mechanism to deal with subspace redundancy. But scalability of its visual representation is one major shortcoming. The target audience for this system are the users expert in data mining and visualization.

3.3 Morphological Operators based Method

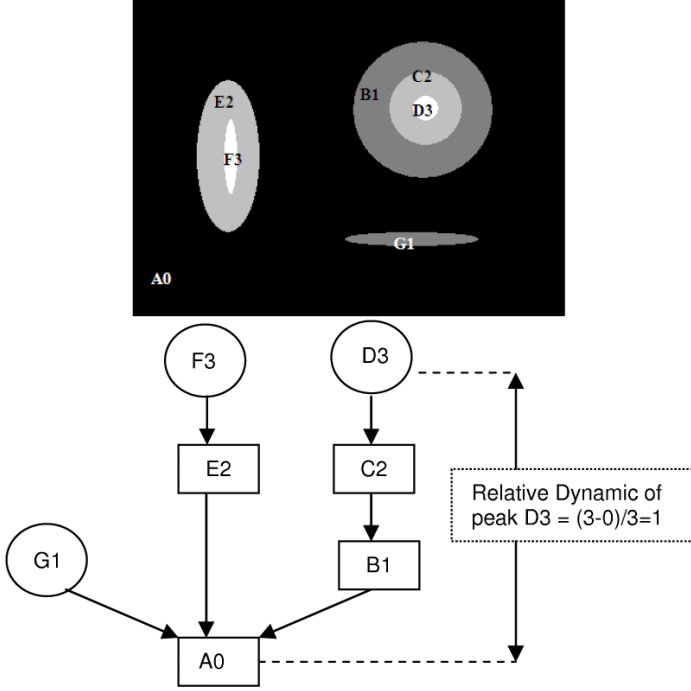


Fig. 4. Connected Flatzones are depicted by circular discs like E2,F3 etc (above) and its corresponding Maxtree is represented (below) (This image is taken from [23])

In addition to ranking subspace for clustering, this novel method also reveals other important quantitative information related to a subspace like the number of clusters present, the degree of separation and significance of clusters. The idea behind this approach is that subspace is considered relevant if it has multiple Gaussian distribution spreads over the attribute range. It takes high dimensional data as input, which is then transformed from parametric space to discrete image space in the form of grid-based density field. This is done by estimating the density of each subspace using kernel density estimation [24]. Morphological operators [25] [26] are applied to this density field to extract connected components called flat zones as demonstrated in Fig.4(above) (where flat zones are represented by circular discs e.g. E2, F3 etc). Morphological operators create a hierarchy of flat zones which is represented using Maxtree data structure [23] as demonstrated in Fig.4(below). In Maxtree [27] [28] root corresponds to lowest intensity flat zone and leaves represent highest intensity flat zones. This representation further facilitates subspace analysis as clusters are indicated by high density regions. Therefore, the number of clusters in subspace is equal to the number of leaves in Maxtree representation. Next, meaningful subspaces for clustering are filtered by applying quality criteria based on dynamics of local maxima of density. Local density maxima are calculated using Maxtree representation of density image. The dynamics of a local maximum is the difference between the intensity value of the corresponding leaf and the intensity value of its first ancestor with multiple children that corresponds to the deepest minimum in the neighborhood.

The advantage of this method is that it allows human

interaction i.e users can interact with the system during the computation of quality measure and can aid to improve the results. It reveals additional information related to a subspace e.g. the number of clusters, significance of clusters etc. Also, this method is scalable and is capable to cope with the enormous amount of data. This method also contributed some visualizations which facilitates in-depth exploration of ranked subspaces. These are line graphs (supporting visualization of 1-dimensional subspaces), scatter plots (supporting visualization of 2-dimensional subspaces) and PCA scatter plot projections for higher dimensional subspaces.

3.4 Entropy-based Method

This work proposes new meaningful criteria to prune uninteresting subspaces which do not incorporate strong clustering. The objective is to ascertain the subspaces having "good clustering". This work suggests three criteria to define goodness of clustering in subspaces. These are:

- *High Coverage*: Fraction of database covered by the dense unit. Subspace having prominent clusters will have high coverage.
- *High Density*: This is another additional criterion that can be used along with Coverage. For instance, the dense units identified based on coverage having uniformly distributed data points is less qualified as a cluster than another dense unit with same coverage having densely packed points in the small region.
- *Correlation of dimensions*: Although the subspace may contain clusters, it may not be intriguing if there is no correlation between the dimensions of the subspace. This criterion states that interesting subspaces should not have completely independent dimensions, however, they might not have a very strong correlation.

This paper proves that all the above-stated criteria can be measured simultaneously using *Entropy* from information theory [29]. Entropy is the measurement of disorder in the dataset or uncertainty of random variables. Intuitively uncertainty (Entropy) decreases with clustering. This is because, when points are clustered, then a particular point is likely to fall within a certain area. Therefore subspace with clusters has lower entropy than subspace without clusters having uniformly distributed data points. To reduce the search space, the algorithm uses a bottom-up approach and iterates in an apriori fashion to find candidate subspaces based on threshold values ω of entropy. A subspace with entropy below ω is considered to have good clustering. ENCLUS defines a candidate set of subspaces by coupling downward closure property on entropy and upward closure property on dimensional correlation (which is also detected using entropy). The outcome of this algorithm is a subset of subspaces with good clustering and minimally correlated dimensions.

3.5 Representative Factor based Method

This work proposes a method i.e *Representative factor* [30] which is tailored to present subgroup of dimensions. It has the ability to reveal relationships between dimensions and thus can boost the reliability of high dimensional data analysis process. Representative factor (denoted by D_r) is

derived dimension from subset of dimensions denoted by D' (where $D' \in 2^D$). Depending on the nature of data set under consideration the mapping of $D' \rightarrow D_r$ can be computed using three different methods i.e.

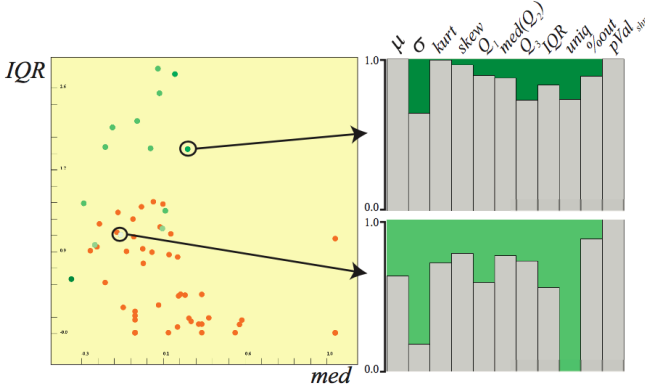


Fig. 5. Illustration of profile plots to visually analyze, compare and evaluate representative factors along with the original dimensions. (This image is taken from [30])

- **Projection Factor:** It is suitable to represent highly correlated dimensions. Firstly to identify correlated dimensions, MDS projection whose distance matrix is computed using the correlation between dimensions is used. This MDS projection places highly correlated dimensions together. The correlated group of dimensions can be extracted from MDS projection and feature reduction method like PCA is used to compute representative factor for these dimensions.
- **Distribution Model Factors:** It is suitable to represent a subgroup of dimensions with similar underlying data distributions. Such subgroup of dimensions can be selected using Kurt vs skew scatter plot on dimensions. D_r in this method is modeled distribution, derived from distributions of the selected group of dimensions.
- **Medoid Factors:** It is suitable to represent the redundant group of dimensions or dimensions with same contextual properties. The user can select one of the dimensions in D' as a representative factor.

This work also suggests a method i.e. profile plots to visually analyze, compare and evaluate representative factors along with the original dimensions. Representative factor tries to best convey characteristics or statistical properties of represented dimensions but it does not represent all properties equally. Profile plot visualization facilitates evaluation of the difference in information between representative factor D_r and associated dimensions D' . The difference of information is normalized in the range of 0 to 1. The higher value (close to 1) of statistical property in profile plot represents higher similarity of information between D_r and D' . Profile plots are demonstrated in Fig.5 where two factors are evaluated. The first factor better represents most of the properties of underlying dimensions as compared to the second factor.

The advantage of this method is that it facilitates structure-aware analysis of the high dimensional dataset and iden-

tifies local domains where computational tools can be applied. Therefore, it can be integrated into working pipeline of these tools for understanding high dimensional data. However, to adopt this approach users are required to have a deep understanding of statistics and the computational tools to analyze high dimensional data. Thus making its learning curve steep.

4 INFORMATION VISUALIZATION PHASE

In order to understand high dimensional data, it is important to understand its clusters, outliers, and correlations. To accomplish this task, there is a necessity to develop a system which encourages effortless comparison between subspace patterns. Subspace clustering has introduced a new paradigm to understand and analyze high dimensional data by localizing search to few dimensions which uncover hidden patterns. However, patterns might not be similar across subspaces. Thus posing challenge to identify, interpret and compare patterns in subspaces. Visual analytics plays a very critical role in comprehending high dimensional data patterns. But, the challenges faced are:

- Handling enormous number of candidate subspaces and possibly redundant cluster results.
- Effectively visualizing diverse characteristics of subspace clusters like size, shape, dimensions included etc.

Some visualization techniques [31] [32] [33] [34] [35] can analyze and interpret clusters but are not desirable for high dimensional data as their scope is limited to full space clustering results. Therefore, recently many visualizations have been proposed to achieve the desired interpretation of high dimensional data using subspace clusters. Based on the type of information they reveal, we have categorized them as follows:

- Visualizations Analyzing Subspace Clusters
- Visualizations Analyzing High dimensional space

This section discusses some of the visualizations in each category.

4.1 Visualizations Analyzing Subspace Clusters

4.1.1 VISA

Visual subspace clustering analysis (VISA) [36] gives novel visualizations to analyze and visualize subspace cluster characteristics in depth. VISA is one of the very initial and important works which proposes both in-depth and global visualizations that help in the interpretation of subspace cluster results.

Density in different subspaces with different dimensionality is not comparable. Unlike most of the existing density based subspace clustering algorithms which ignore this effect and compare incomparable density values of different subspaces with a fixed threshold, Dimensionality Unbiased Subspace Clustering algorithm (DUSC) [37] eliminates dimensionality bias and leads to comparable clustering results. Comparable clustering results are required to generate reliable visualizations. Therefore, DUSC is used in this approach as subspace clustering method to obtain

clusters $\{(c_1, s_1)(c_2, s_2)....(c_n, s_n)\}$ where c_i is cluster and s_i is subspace associated with it. Resulting visualization of VISA focus on following:

- **Subspace overlap:** i.e dimension overlap for subspaces under consideration.
- **Object overlap:** i.e common objects between subspace clusters.
- **Interestingness of a cluster:** Interestingness is based on expected density in a subspace.

This paper also proposes a novel distance function which allow comparison of subspace clusters. It takes into consideration both subspace overlap (where S_i and S_j are subspaces) and cluster object overlap (where C_i and C_j are clusters in subspaces S_i and S_j respectively).

$$\beta(1 - \frac{|S_i \cap S_j|}{|S_i \cup S_j|}) + (1 - \beta)(1 - \frac{|C_i \cap C_j|}{\min\{|C_i|, |C_j|\}})$$

Subspace overlap or object overlap can be favoured by adjusting the β value.

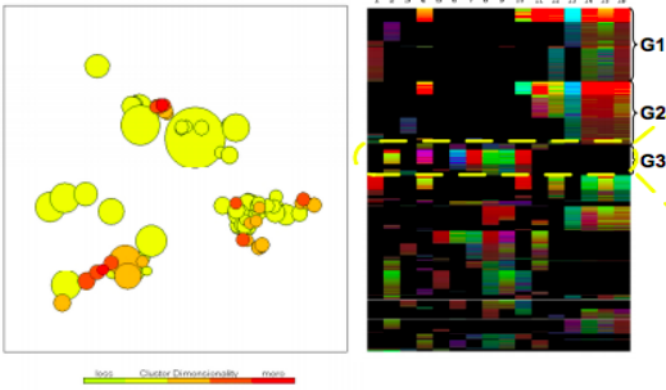


Fig. 6. Illustration of MDS Visualization of VISA (on left) and Matrix Visualization of VISA (on right) where G1, G2 and G3 represents groups of subspace clusters (This image is taken from [36])

Multidimensional scaling (MDS) is a dimensionality reduction technique which tries to preserve pairwise distance of objects in the dataset. Therefore, MDS projection based on above distance measure is used to generate an overview of all distances between subspace clusters. As shown in left side of Fig.6, this visualization consists of circular representation of subspace clusters, where the size of the cluster is visually encoded by radius and subspace is encoded by color. This visualization is made interactive to facilitate in-depth analysis of subspace clusters. In MDS image, similar subspace clusters form groups. A group is defined as a set of subspace clusters that are similar to subspace cluster of high interest (i.e anchor of the group) or in other words set of subspace cluster which are at d distance from the anchor subspace cluster. Matrix representation (shown in right side of Fig.6) is used to visualize the groups of subspace clusters where objects represent rows and columns represent dimensions. Colors are used to represent the dimensions that are part of subspace. If a dimension is not part of the cluster then it is represented by black. Objects are ordered according to decreasing interest. By pointing to individual object additional information about the cluster like interestingness, dimensions included etc. can be visualized. And

overlapping subspaces can also be visualized very clearly from this visualization. The drawbacks of this visualization are as follows:

- Overlap between clusters in MDS projection often cause cluttered visualization which is hard to interpret
- Using circles in MDS projection can be problematic as the distance between clusters can be dubious because of the radius of circles
- It does not visualize cluster overlap and subspace overlap clearly. This is because the distance measure is based on both subspace overlap and object overlap. Thus, by observing MDS Visualization one cannot conclude the proportion of subspace overlap and object overlap.

Despite these drawbacks, VISA is a pioneering work as it gives very powerful and compact visualization for subspace clusters. Also, so many other subspace visualizations are based on VISA technique (e.g. S-ADVISER).

Morpheus [38] is an interactive visual exploration tool for

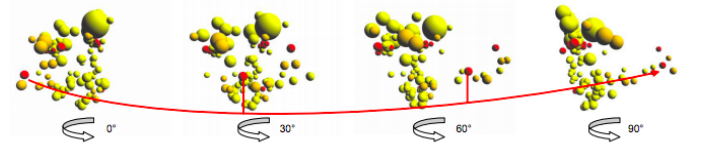


Fig. 7. Detailed 3-d view of subspace clustering results in Morpheus (This image is taken from [38])

visualizing subspace clusters and is based on VISA. Using Morpheus, subspace clustering results can be visualized in 2-d and 3-d setup. 2-d visualization is static. On the other hand, 3-d visualization is dynamic and facilitates interactive exploration of subspace clustering results. Morpheus 3-d visualization helps to overcome the drawback of VISA i.e overlapping clusters causing cluttered visualization. This is demonstrated in Fig.7

4.1.2 ClustNails

It is a interactive visual subspace cluster analysis system that integrates novel visualization techniques for mining subspace clusters and analyzing characteristic properties. Moreover various interaction techniques have been integrated in order to support navigation and interpretation of cluster results.

It take high dimensional data as input, then user selects subspace clustering algorithm from set of pre-provided algorithms from OpenSubspace library of WEKA [40]. High dimensional data is clustered based on subspace clustering algorithm selected by the user (e.g. Clique, Proclus, Doc etc.) and the results are displayed using two visual representations (1) Spikes and (2) HeatNail [39]. These visualizations support navigation and interpretation of subspace clustering results. Spike view is a cluster oriented view where cluster is visualized using circular area containing radial spikes as shown in Fig.8. Size of the circle represents number of records contained in cluster and each spike correspond to a dimension of subspace in which cluster is formed and its length represents the importance of that dimension in that

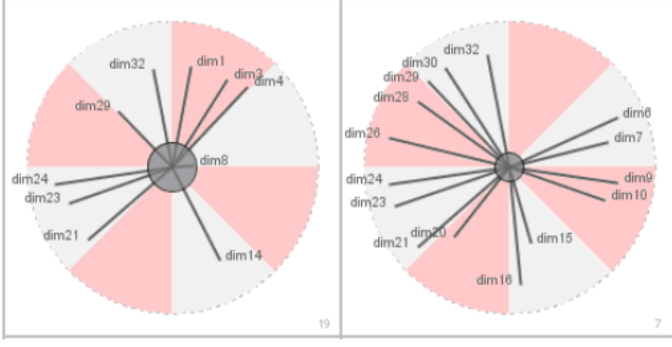


Fig. 8. Spike View visualization of ClustNails (This image is taken from [39])

cluster. The order of the dimensions is identical in spike view for each cluster, this arrangement allow to identify overlapping dimensions in cluster comparison.

HeatNail is a record centric view which extended heat-map

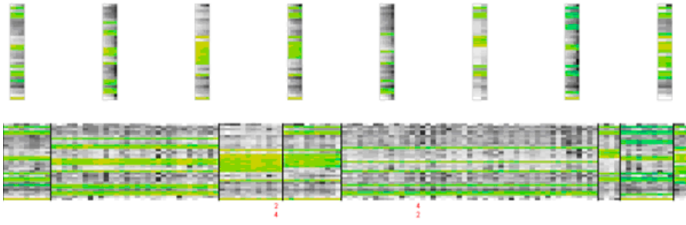


Fig. 9. HeatNail visualization of ClustNails (This image is taken from [39])

with rows representing dimensions and columns representing records or data items. It is demonstrated in Fig.9. Data items are grouped based on clusters and clusters are aligned side by side separated by a line. This visualization allows users to perceive distribution of data by displaying data values across dimensions.

Both the visualizations of ClustNails are understandable and uncluttered. The advantages of ClustNails can be stated as follows:

- **Allows cluster comparison:** ClustNails introduces ordering on dimensions, records, values and subspace clusters which facilitates detection and analysis of (dis)similarity between subspace clusters.
- **Clearly reveal properties of individual cluster:** like records and dimensions involved, weight of each dimension etc.

In contrast to the advantages stated above there are certain shortcomings to this visualization.

- It does not visualize overlapping in terms of cluster objects.
- As the size and dimensionality of input data increases. It suffers from scalability issues.
- There is no way to visualize quality of clusters.

4.1.3 S-ADViSED

S-ADViSED [41] is a visual framework which seeks for exploration and validation of subspace clustering results. It assists Epidemiologists investigating factors contributing

towards the outbreak of diseases. For this, it attempts to comprehend sub-populations, sharing determinant factors for diseases. Sub-populations are determined by subspace clustering. S-ADViSED combines several visualizations to accomplish Epidemiologists requirements. Epidemiology data is a multidimensional data and its dimensions can be broadly categorized as (I) **Habits:** Data obtained by interviewing (e.g smoking), (II) **Physical conditions:** Data obtained by physical examination (e.g blood pressure), (III) **Laboratory tests results,** and (IV) **Medicaments.** Epidemiologists are generally interested in

- **R1:** Low dimensional subspace clusters to avoid over-fitting.
- **R2:** Sufficiently large cluster size to support evidence.
- **R3:** In-depth information i.e understandable and crisp visualization of dimensions associated (and not with) subspace cluster.
- **R4:** Object and dimension overlapping in subspace clusters.
- **R5:** Comparison with global mean as it is necessary to compare subgroups with the whole population.

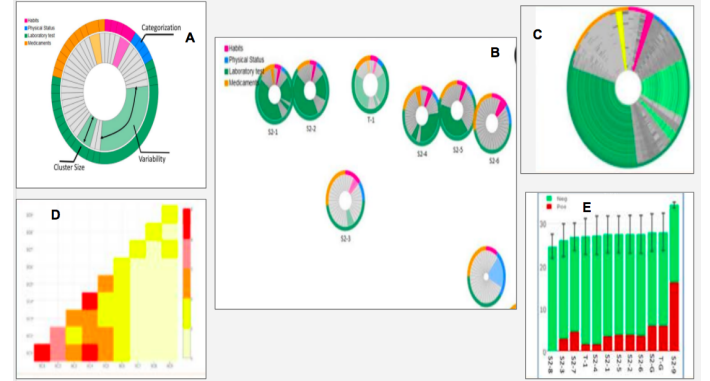


Fig. 10. (A)Subspace cluster is illustrated by Donut chart where radius of Donut represents cluster size, sector represents dimensions and color is assigned based on categorization of dimensions, (B)MDS projection of subspace clusters which gives overview of all distances between subspace clusters where distance is based on object similarity and dimension similarity, (C)Donut heatmap, (D)Heatmap, and (E)Stacked errorbar(This image is taken from [41])

Therefore, S-ADViSED attempts to yield the global view of subspace clusters using **Donut charts** illustrated in Fig.10(A) where sectors represent dimensions (colored ones are involved in subspace cluster and the color is assigned based on the categorization of dimensions as stated before), radius depicts the size of cluster and size of the sector represents variance in dimension. This visualization supports R3. These Donut charts representing subspace clusters are then projected on 2-d space using **MDS projection** as in VISA (demonstrated in Fig.10(B)) supporting R4. In addition to this visualization it also provide some other visualizations like **Donut heatmaps** (Fig.10(C)), **heatmaps** [42] (Fig.10(D)), **Stacked errorbars** (Fig.10(E)) etc. Donut heatmaps steer to further analysis of subspace cluster as each ring in Donut represents an individual object of the cluster and its values are mapped to color, darker color represents greater value whereas brighter color represents smaller value. Heatmap is $n \times n$ matrix, where n represents the number of subspace

cluster. Each cell of this matrix represents similarity between two subspace clusters, fulfilling R4. Stacked errorbars compare subspace clusters with the global mean of whole data accomplishing R5.

The advantage of this approach over others is that it contributes a clear visualization to compare and analyze variance in dimensions through Donut sector size.

4.1.4 SubVIS

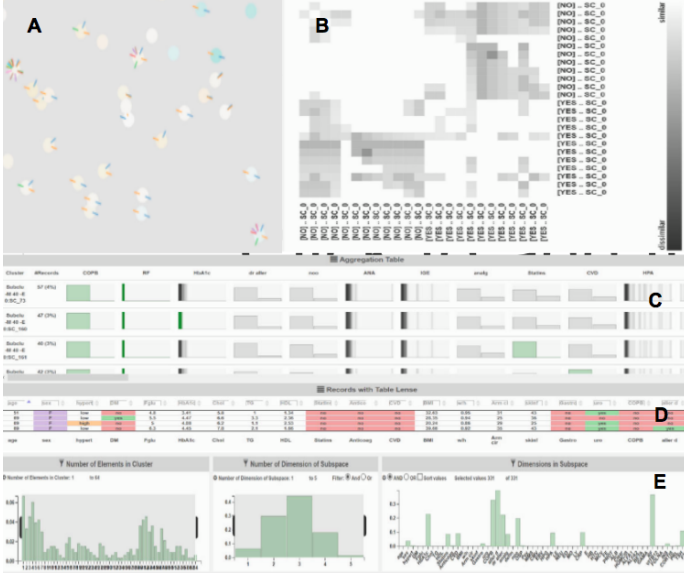


Fig. 11. Different visualization in SubVIS Tool including (A) MDS projection, (B) Heatmap Visualization, (C) Aggregation table Visualization, (D) Table Lens, and (E) Barchart Visualization (This image is taken from [43])

SubVIS [43] is an interactive visual tool to explore and analyze subspace clusters comprehensively. This work is based on the previous work [44] which analyze subspace clusters in bio-medical data [45]. It integrates both subspace analysis algorithm and interactive visualization to tackle challenges confronted by subspace clustering algorithms like Interpreting complex subspace cluster results, redundancy in clusters etc. In contrast to VISA which emphasis on global comparison of subspace clusters, SubVIS along with global overview also capacitate users to study individual clusters characteristics.

SubVIS follows Shneiderman's information seeking mantra [46], i.e overview and then details on demand. Therefore, work-flow of SubVIS pursues with a general overview of subspace clusters similarity using MDS projections based on different distance measures that can be selected by users interactively. These distance measures can display different aspects of subspace clusters like subspace overlap, cluster member overlap etc. One of the MDS projection is demonstrated in (Fig.11(A)). Furthermore, to explore details of subspace cluster similarity heatmap visualization (Fig.11(B)) is introduced. In this visualization, the brightness of each cell indicates pairwise cluster similarity. Overview of characteristics like distribution of dimensions and size within selected subspace clusters can be obtained by bar chart representation (Fig.11(E)) displaying properties of subspace clusters. Next level of exploration focus on zooming and

filtering. A user can select subspace clusters of interest from MDS projection and can get aggregated view of cluster members from aggregation tables (Fig.11(C)). In aggregation, table rows represent clusters and columns indicate dimensions (dimensions included in the cluster are represented by green color) and each cell indicate aggregated data record within the cluster. Dark color represents high occurrence. Last level of exploration focus on details on demand where users can explore actual data records using table lens [47] view (Fig.11(D)). This recent work has some very new features as compared to previous subspace cluster analysis approaches [39] [36]:

- **Identification of redundant subspace clusters:** Closer inspection of the heatmap, MDS projection and bar charts can reveal redundancy in subspace clusters.
- **Filters:** Filtering mechanism is introduced to facilitate selection of clusters for further analysis. This is based on various aspects like number of dimensions, number of cluster objects, dimensions of interest etc.
- **Multiple views of same high dimensional data:** SubVIS integrates a wide variety of distance measures upon which MDS projections can be built interactively. Multiple MDS projections along with their heat map visualizations are paramount to explain different views on same high dimensional data (including subspace and object overlap).

4.1.5 Dimension Re-Construction

The potential aim of this approach is to reconstruct new subspace to preserve interesting cluster information, using original subspace projections [48]. Reconstructed subspace projections facilitate users to perceive subspace clusters that were not prominent in original subspace projections. Mechanically merging subspaces can lead to cluttered visualization of subspace clusters because of multiplied number of dimensions. This calls for the generation of a manageable number of new informative dimensions which could be merged with original subspaces to preserve interesting cluster information. It defines interactive visual analytic pipeline through which new reconstructed dimensions can be added to subspaces to improve cluster results and track clusters in different subspaces.

Process of dimensional reconstruction is composed of following steps: (I) **Direct Data Projection:** It takes high dimensional data as input and performs 2-d projection on it using projection techniques like PCA, MDS etc. which make data visually accessible for analysis, (II) **Subspace Analysis:** It captures 2-d projections of all subspaces of high dimensional data and performs interactive cluster analysis in different subspaces, and (III) **Dimension Reconstruction:** Lastly, it generates new dimension from 2-d projections of subspaces which incorporate interesting cluster information. New subspaces are created by integrating new dimensions with original subspaces. This may improve clustering results.

This work-flow is illustrated in Fig.12 which intakes categorical high dimensional data (consisting of three categories represented by red, blue and green color). Fig.12(a) demonstrates 2-d projection of full dimensional space, Fig.12(b)

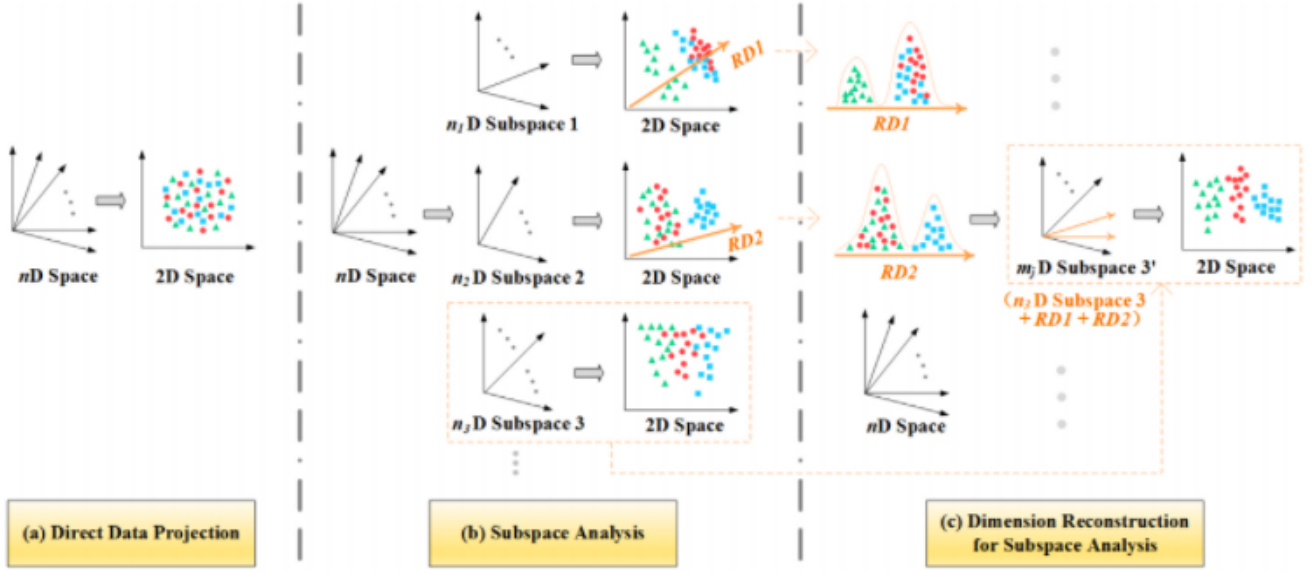


Fig. 12. Dimension reconstruction for visual interactive subspace cluster analysis. (This image is taken from [48])

demonstrates three subspaces. It can be observed that green and blue samples are separable in *subspace 1* and *subspace 2* respectively. In *subspace 3* green and blue samples are again separable but red samples which lies between two, mingle with the boundary points of green and blue cluster. *RD1* and *RD2* are reconstructed dimensions in *subspace 1* and *subspace 2* respectively, which clearly separates green and blue sample respectively when data is projected on them. Merging *RD1* and *RD2* with *subspace 3* leads to clear cluster representation in new subspace projection.

Although this technique does not explore subspace cluster characteristic properties (as in ClustNails [39], SubVIS [43] etc.) but it has advantages over other techniques as:

- It is one of the very few works that focus on the construction of new subspaces interactively by integrating new informative dimensions and original dimensions.
- User can visualize and analyze subspace clusters in reconstructed subspace which were not prominent in original subspaces. Subspace clusters visualization in new subspaces helps in comprehending high dimensional data.
- It can be considered as alternative dimensional reduction technique. Thus can be beneficial to users in multiple real-world applications.

On the other hand, the limitations associated with this technique are:

- **Usability:** It is a complicated system, involving human interactions to construct new subspaces, So it suffers from usability issues. However, it will be more suitable if users are already aware of some interesting subspaces right at the beginning.
- **Scalability:** Number of subspaces increases exponentially with dimensions. Therefore, finding subspace projection for all subspaces is a time-consuming task.

- **Loss of information:** Data becomes distorted when projected in 2-dimensional space from higher dimensional space. This results in loss of information and might lead to wrong conclusions about data.
- Lastly, reconstructed dimensions have no physical meaning as original dimensions. This can further complicate interpretation task.

4.1.6 SUBEVAL

Assessing quality of clustering is a very difficult task as different clustering algorithms are based on different methodologies. The central goal of this evaluation is to see if clustering results make sense or not. Researchers till date, generally use statistic measures for the purpose of analysis of subspace clustering quality. SUBEVAL [49] is a pioneer work that provides a novel method for evaluation of the quality of subspace clusters through visual support. This approach attempts to access the quality of subspace clusters at different granularities such as non-redundancy, compactness, and the dimensionality of clusters. The main challenges in accessing subspace cluster quality are as follows:

- Different clusters may exist in one subspace.
- Similar clusters may exist in different subspaces.
- Similar clusters may exist in similar subspaces.

This approach defines three criteria as a baseline to visually access the quality of subspace clusters.

- **Non-Redundancy Criteria (C1):** Clusters should not be redundant in terms of cluster members and dimensions involved in subspaces of respective clusters. Moreover, similar clusters occurring in different, non-redundant subspaces (called *Cluster-Splitup* in Subspaces) should be avoided.
- **Object and Dimension Coverage Criteria (C2):** Practically a subspace clustering algorithm is used,

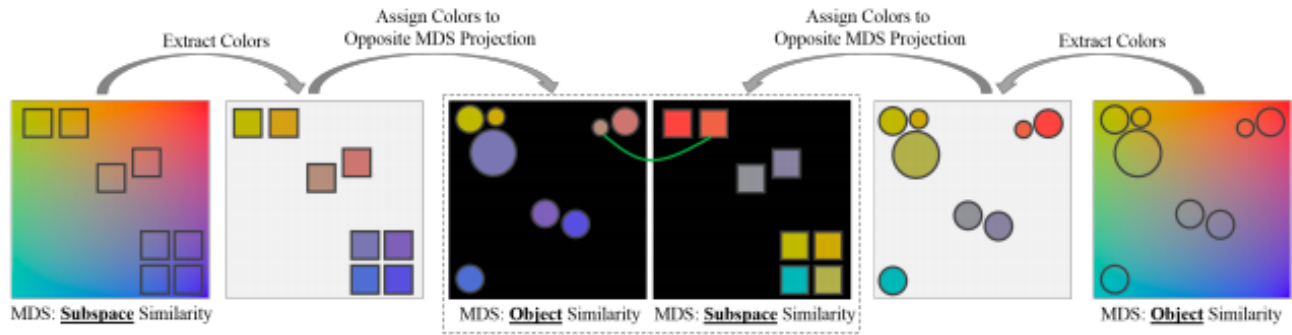


Fig. 13. Two Interlinked MDS projection are illustrated, one from subspace similarity based MDS projection to object similarity based MDS projection and other vice-verse. Here square shape is used for visual encoding of subspace clusters in MDS projection based on subspace similarity and circular shape is used for visual encoding of subspace clusters in MDS projection based on object similarity. It demonstrated the work-flow of assigning colors from one MDS projection to other.(This image is taken from [49])

if it showcases high coverage of both objects and dimensions. This helps to explore the global patterns in the high dimensional data.

- **Cluster Characteristics Criteria (C3):**
Subspace clustering algorithms should try to optimizing cluster characteristics like compactness, separability, cluster size etc.

The Conceptual illustration of this method is presented in Fig.13 where two interlinked MDS projections are used to understand redundancy aspect of subspace clusters. One MDS projection is based on *cluster object similarity* and other is based on *subspace similarity*. The basic idea is to extract colors from subspace clusters in MDS projection (based on subspace similarity) and assign the extracted cluster colors to corresponding clusters in MDS projection based on object similarity (This whole procedure can be executed other way round i.e. starting from MDS projection based on object similarity). This combines both similarities together and visualizes pair wise similarity of all clusters. Subspace clusters are **truly redundant** if they exhibit similarity both in terms of cluster membership and dimensions associated with their subspaces. Lets elaborate interlinked MDS projection which assigns colors extracted from subspace similarity based MDS projection to object similarity based MDS projection in Fig.13. Subspace clusters being close to each other in interlinked MDS projection and sharing similar color are truly redundant. On the other hand, clusters which are close to each other, but sharing different colors exhibits object similarity but are different in their subspace topology. Similarly, interlinked MDS projection from object similarity based MDS projection to subspace similarity based MDS projection can be interpreted.

Based on the above-obtained visualization, one can evaluate the quality of subspace clusters and can decide if subspace clustering algorithm is good or bad. It gives an overview of redundancy aspect by visualizing cluster similarity and dimension similarity. Jaccard Index [50] are used to intuitively compute the similarity between two clusters or subspaces. Further interpretation of this visualization is summarized in Fig.14. SUBEVAL also access quality criteria *c2* and *c3* using some other visualizations like object and subspace coverage bar charts, Cluster size distribution histograms, enhanced

Interlinked MDS Projection	Interpretation
	Subspace Clusters are Highly Redundant
	Subspace Clusters exhibits Low Redundancy
	Clusters with many members or dimensions are represented by large glyphs in the MDS plots. There is a possibility that these are Non Compact clusters .
	Too many small clusters do not provide good generalization of data. These are Small splinter clusters .
	Largely overlapping clusters or Cluster Split Up in Subspaces do not imply redundancy but creates a suspect that these objects form cluster in high dimension

Fig. 14. Different approaches for High Dimensional Data Analysis(This image is taken from [49])

parallel coordinate system etc.

This tool is useful as users can select various subspace clustering algorithm, vary its parameters and can even evaluate new algorithm results against original assumptions.

4.1.7 BEADS

BEADS [51] system uses 2 or 3-dimensional shapes to build or describe shape and size of higher dimensional data clusters. The input to this system is already computed high dimensional cluster. For each high dimensional cluster, it focuses on representing its shape in 2-dimensional space. The subset of points whose shape can be identified through

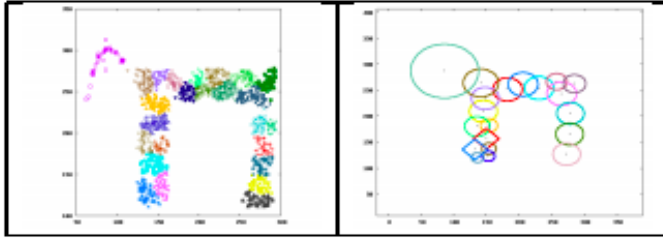


Fig. 15. Beads System, identifying shape and structure of chameleon dataset (this image is taken from [51])

mathematical framework is called BEADS. It uses the *break and build* approach where the cluster is first broken into smaller pieces, then 2-d shape that best fit each piece is identified (e.g. circle, square etc.) which is further used as building block for building the overall structure of cluster. Fig.15 demonstrates the applicability of BEADS on CHAMELEON dataset [52], where BEADS plot closely resembles original cluster shape prevailing in the dataset. The central idea behind this system is that high dimensional cluster shape is the composition of small primitive shapes. The advantage of beads system is that it has the ability to handle complex and random clusters shapes of high dimensional data. Clusters are represented by spatially connected 2-d beads.

This section on *subspace cluster analysis* is summarized by Table.1 which draws a clear comparison between above-stated methods against their ability to analyze different characteristics of subspace clusters.

4.2 Visualizations Analyzing High dimensional space

Humans have a hard time in understanding and navigating the world of high dimensional data. Recently researchers have paid attention to understand the orientation of high dimensional space. This section focuses on recent advancements that enable users to gain a true appreciation of high dimensional space.

4.2.1 TripAdvisor

This work [53] attempts to draw a comparison between high dimensional space navigation and sightseeing trip. This interactive tool facilitates global navigation of maps (i.e. high dimensional space) containing sites of interest (i.e. Subspaces having clusters) and local exhaustive and detailed exploration of sites (i.e. data in a particular subspace). It is very hard to navigate a world of high dimensionality. However, this system grants remarkable flexibility in terms of high dimensional space navigation and orientation by introducing a touchpad interface which further aids the perception of multivariate relationships and cluster analysis assisted by exploiting motion parallax. It has two applications which are: (1) Exploratory analysis of data (2) Cluster analysis. It attempts to perform these two tasks directly in high dimensional space.

This system enables users to gain a true appreciation of high dimensional data space by commencing exploration from interesting sites. The following preprocessing steps are followed to identify these interesting sites or viewpoints:

TABLE 1
Comparison of Subspace Cluster Analysis Methods

USAGE	ClustNails	VISA	S-ADViSED	SubVIS
Overlapping cluster objects	✗	✓	✓	✓
Overlapping cluster dimensions	✓	✓	✓	✓
Dimensions involved in subspace cluster	✓	✓	✓	✓
Ordering technique	✓	✗	✗	✗
Comparing size of cluster	✓	✓	✓	✓
Weight of dimension	✓	✗	✗	✗
Distribution of data in dimension involved	✓	✓	✓	✓
Use of WEKA	✓	✗	✗	✓
Bracketing	✗	✓	✗	✗
Subspace Visualization	✓	✓	✓	✓
Interestingness of subspace cluster	✗	✓	✗	✗
Distribution of data in non involved dimensions	✓	✗	✓	✓
Category of dimension	✗	✗	✓	✗
Relation between attributes	✓	✗	✓	✗

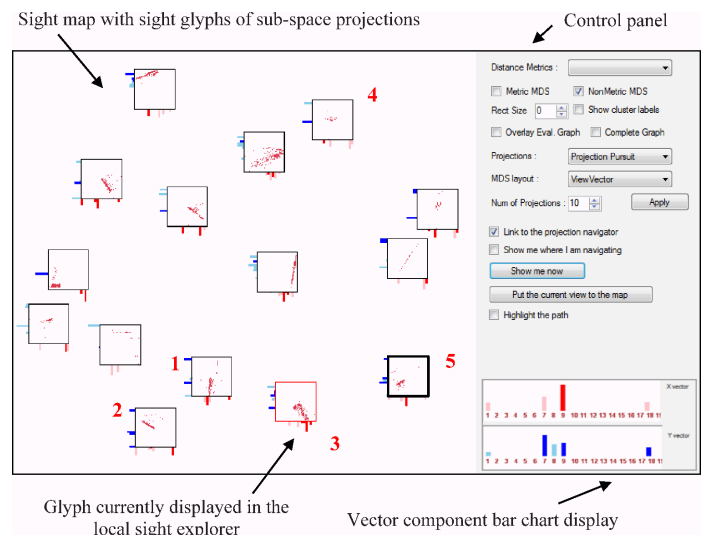


Fig. 16. Global site map with sites or viewpoints of subspace projection shown as glyphs or icons. Glyph labeled by 3, highlighted by red boundary is currently shown in Local site explorer. (This image is taken from [53])

- (1) Use entropy based subspace algorithm i.e. ENCLUS to determine independent subspaces having dense structures.
- (2) Use projection algorithm like PCA on candidate sub-

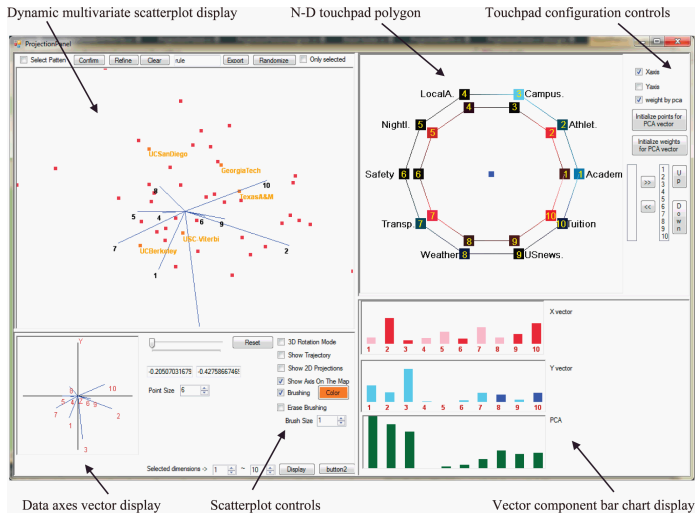


Fig. 17. Local site explorer having four components i.e Dynamic multivariate scatterplot display, Data axis vector display, Polygon touchpad display and Bar chart display (This image is taken from [53])

spaces to determine interesting viewpoints. These sites are then organized on the **global site map** using MDS, which layouts spatial similarity between sites. Each site on global map has its icon augmented with bar charts along x and y-axis representing the significance of relevant dimensions. The global site map is presented in Fig.16. Clicking on a site in global map triggers detailed projection view of that site in the **local site explorer** (as shown in Fig.18). The local site explorer has 4 major components: (1) *Dynamic multivariate scatterplot display*- It projects standard biplot [54], (2) *Data axis vector display*- It envisions the projection and labels of the dimension axis vectors, (3) *Polygon touchpad interface*- It enables comprehensive insight into data space by facilitating navigation in biplot environment, and (4) *Bar chart display*- It projects three bar charts. One to visualize PCA spectrum representing the strength of each PCA vector. Others, visualize weight of each dimension along x and y-axis of biplot.

The advantages of this framework over others are:

- It provides a platform where important exploration tasks like clustering can be performed directly in high dimensional space using motion dynamics which can be very helpful in exposing interesting patterns in multidimensional space [55]. It aids users to examine motion trials in biplots visually and brush them to discover similar patterns. As a result, users can appreciate clusters that do not fit the assumptions of automatic clustering algorithms.
- It introduces the notion of navigation in biplots. Thus making biplots much more efficient in conveying multivariate relationships.

The only major shortcoming of this framework is that the system is prone to scalability issues as it comprises of components like N-D touchpad polygon, biplots.

4.2.2 Subspace Voyager

It is another map-based system dedicated to organize and manage subspaces to comprehend high dimensional data.

It is challenging to analyze the structure of high dimensional data using conventional multivariate scatter-plots. Subspace voyager [56] solves this problem as it attempts to explore high dimensional data space by decomposing it into the continuum of generalized 3D space. This system is motivated from TripAdvisor as it uses biplot visualization. To analyze, explore and navigate continuum of 3D subspaces, augmented trackball interface is used which is an advancement of navigation pad interface in TripAdvisor. Analysts can access the relevance of each dimension during visual interaction with trackball interface from the size and opacity of dimension labels plotted at the periphery of the display. Subspace voyager has three major components as demonstrated in Fig.14:

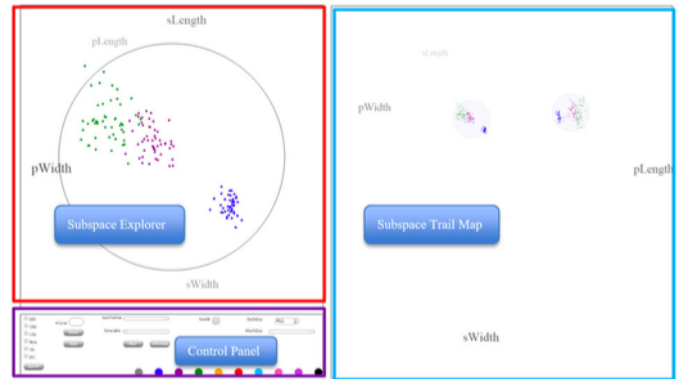


Fig. 18. Subspace Voyager interface is demonstrated having three components: the control panel, the Subspace Explorer (SE), and the Subspace Trail Map (STM). The control panel allows users to set the various parameters of the system. The SE is coupled with the trackball interface and labels placed at the periphery of circular boundary. The size and opacity of labels of dimension represent their influence on the display. The SE offers various interactions for users to examine the data. The STM holds a set of interesting views selected by users. (This image is taken from [56])

- Control panel: It allows users to set various parameters of the system. It can be used to generate 3D subspaces which can be further analyzed.
- Subspace Explorer: Subspace data is projected and displayed in 3D subspace explorer trackball with Ant Colony Optimization[] that automatically generates best trackball configuration. Users can interact with this interface to explore current subspace. It also facilitates the transition from one generalized 3D subspace to another based on dimensional weight. Users can analyze and chase clusters by brushing them into different colors.
- Subspace Trial Map: Users can keep track of interesting findings (discovered during exploration process) by saving Subspace Explorer view in subspace trial map. These views can then be connected by lines to construct animated tours.

When data is projected from higher dimensional space to lower dimensional space, there is a loss of information which can further lead to ambiguities. One of the advantages of this work is that it overcomes such ambiguities. This is accomplished by using motion parallax in projections (as in tripadvisor[], ggobi[] etc).

4.2.3 Heidi

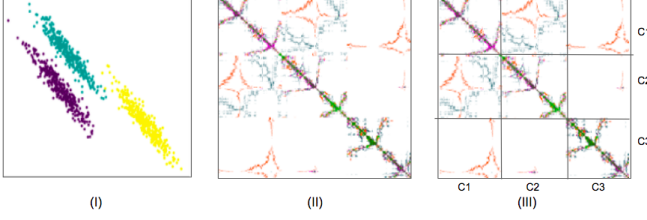


Fig. 19. (I) Shows 2-dimensional synthetic dataset with three clusters C1,C2 and C3 represented by yellow, green and purple color coding respectively, (II) Illustrates Heidi Image on this dataset, and (III) Demonstrates cluster wise ordering of points (along rows and columns) in Heidi Image.

Heidi Matrix [57] visualization helps in understanding multivariate data [58], as it captures and conceptually visualizes the closeness of points across various subspaces. Fig.19(II) shows Heidi image on the synthetic dataset. It is a 2-dimensional matrix of size $n \times n$ where n is size of dataset. Each cell (p, q) in this matrix represents knn relationship [59] (i.e whether q is in k nearest neighbour subset of point p) across all possible subspaces. This is done by using bit vector whose size is

$$2^d - 1$$

and each bit stands for a subspace. The least significant bit in bit vector corresponds to lower subspace and most significant bit represents higher subspace. Bit value is 1 if $k-nn$ relationship exists in a particular subspace under consideration else set to zero. The integer value of the bit vector is assigned a unique color.

The procedure of generating Heidi matrix comprises a clustering algorithm that generates K clusters for a given high-dimensional data. And enforces ordering of points, cluster-wise (points belong to a cluster comes together) along the rows and columns which facilitates better visualization and understanding the internal structures of multivariate data. The demonstration of point ordering can be seen in Fig.19(III).

The advantages of Heidi Matrix representation over other visualizations are:

- It presents the complete information or snapshot of whole input data, as it visualizes proximity of every pair of point in dataset across various subspaces by building $n \times n$ matrix based on their $k-nn$ relationship.
- Patterns in this visualization represents the overlap of clusters in different subspaces. Therefore, It aids the inferential process of accessing inter and intra-cluster relationships across various subspaces.
- Since each integer value of bit vector is mapped to a unique color. Therefore, the most prominent color in this visualization can reveal important subspaces in multivariate data.
- Pattern size reveals subspace cluster size.

In Contrast to these advantages, Heidi visualization is a complicated system where the abstract visual mapping scheme makes it difficult to interpret results. The time and space complexity of calculating Heidi matrix is $O(2^d \cdot n^2)$.

Therefore, it suffers from scalability issues (for $d \geq 8$). Moreover, only the users specialized in data mining and visualization can make sense out of this system.

4.2.4 Pattern Trails

The main aim of pattern trails [60] is to explain structural changes of patterns or cluster among different subspaces (placed side by side) through linked representations based on operations performed in attribute space. Pattern trails technique follows the following steps:

- **Subspace reduction:** It focuses on finding interesting subspaces using subspace selection algorithms like SURFING [18]. This is important because otherwise, pattern trails visualization becomes cluttered due to limited display areas.
- **Dimensionality reduction:** After obtaining a reduced subset of subspaces, dimensionality reduction techniques are applied to each subspace to make them visually accessible. 2-dimensional projections of higher dimensional subspaces are obtained by using algorithms from dimensionality reduction pursuit like PCA, MDS etc.
- **Grouping and reordering:** For in-depth analysis, it is important to further reduce subspaces to a smaller number of representative once, in order to reduce over-plotting and visual redundancy between projections. So projections are grouped by applying agglomerative clustering on them. Resulting clusters are then replaced by their prototype projections. These new projections are visually represented by all unique attributes included in all subspaces of their parent cluster. Next, to make sense of pattern transition from one subspace to another, subspace projections are reordered based on data-driven projection based similarity.
- Once the meaningful ordering is obtained, scatter-plots of prototype projections are placed **side by side** to trace and interpret pattern changes.
- **Highlight transition of patterns between subspaces:** Data points belonging to the selected pattern are connected across all subspaces using lines. Pattern transition between subspace describes structural changes in patterns.

This system gives liberty to select any pattern for analysis and enables to draw a conclusion based on lines between subspaces. Tracing these lines help in interpreting and analyzing which attribute cause the effective change in pattern. As shown in Fig.21, between two consecutive subspace projections, an (added, removed or replaced) attribute can have a dominating or non-dominating impact. An attribute is dominant and influential if the pattern transition is **changing** i.e if it significantly controls the structure change of subspace pattern and is non-dominant or redundant if the pattern transition is **stable** i.e if the structure of subspace pattern remains same.

Fig.20 is a snapshot of pattern trails visualization which is performed on University Rankings dataset [61] comprising of 7 numerical attributes (i.e overall score, academic reputation, employer reputation, faculty-student ratio, international faculty, international students and citations per fac-

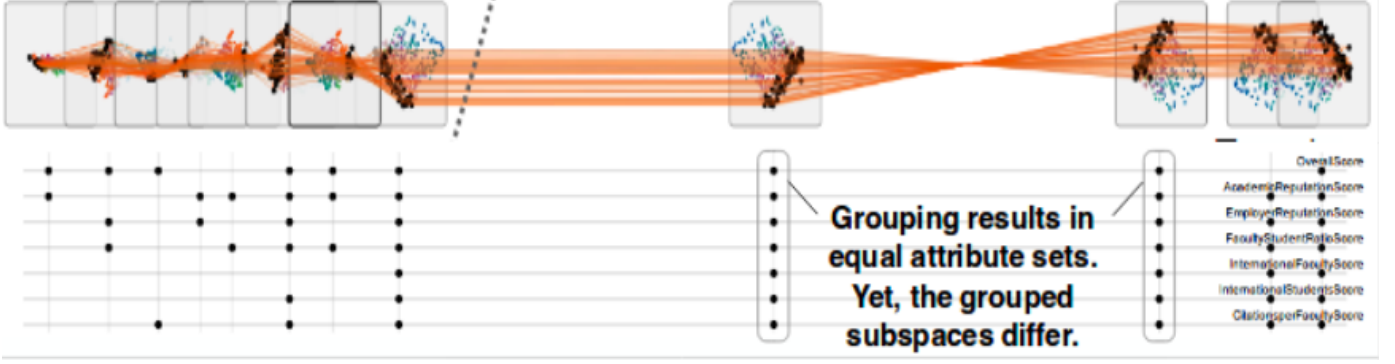


Fig. 20. Illustration of Pattern trails technique on University Rankings Data-set (This image is taken from [60])

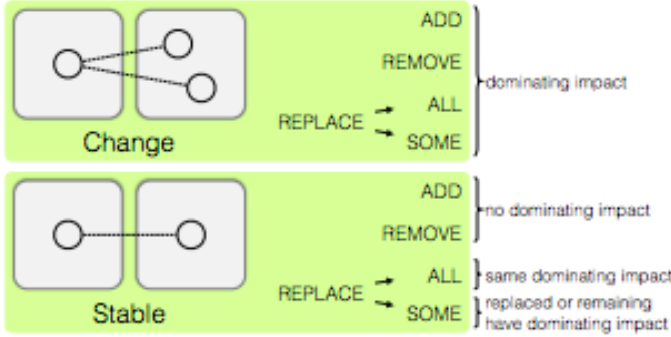


Fig. 21. Illustration of dominating and non dominating impact of attributes based on changing or stable pattern transition respectively in Pattern trails Visualization. (This image is taken from [60])

ulty). It can be observed from the inferential process of pattern trails that last two subspaces differ by attribute *Overall score* and show a static pattern. This means that this attribute has no impact on pattern structure. Further drilling down in this visualization, it can be seen that 9th and 10th subspace projection, add no new insight as **grouping** results in same attribute representation for both subspace prototypes. In 7th and 8th subspace pattern diverges indicate structural change caused by dominating attribute. To find dominating attribute consider the static transition between 6th and 7th subspace, therefore no dominating attribute exists between them. Now, comparing 6th and 8th subspaces, *International Faculty score* is dominating attribute.

The advantages of this method are:

- It contributes meaningful ordering on subspace projections so that change in pattern can be effectively analyzed.
- It gives new insight into high dimensional data by revealing interesting attribute causing a change in pattern structure across subspaces.

The drawbacks of this method are as follows:

- **Scalability:** This work can be applied to at most 8 dimensions. Therefore not scalable.
- **Information loss:** This approach involves applying agglomerative clustering to reduce the number of subspaces. This is done at the expense of information, if more subspaces are combined, the subspace

projection can become less expressive. In order to overcome this shortcoming, one can investigate attributes without using clustering. But the number of subspaces increases exponentially with increase in dimensions and side by side subspace projection visualization in pattern trails can become complicated and difficult to understand. Therefore there is a trade-off between scalability and information loss.

- **Attribute Interpretation:** While comparing pattern transition between two subspaces, the same attribute can be redundant for one pattern and dominant for other. Therefore it is difficult to interpret the global role of the attribute.

5 DISCUSSION

High dimensional data cannot be directly visualized. Therefore, indirect methods are needed to determine insight of data. The grand vision for the future is to support human understanding of high dimensional data. This needs concentrated efforts towards reasoning subjects like:

- What is there in the data?
- Why it is of that way?
- How it can be comprehended (as much as possible)?
- How efficient are the methods involved in comprehending data?

Currently, there are two major challenges faced by the research community while comprehending high dimensional data using subspace clustering methods. These are:

- **Scalability:** Most of the existing techniques to comprehend high dimensional data suffer from scalability issues mainly due to display limitations. These techniques are suitable for multivariate but not for high dimensional data where dimension count is in hundreds or thousands. They fail to provide reliable and interpretable results [20]. Comprehension of multidimensional data where dimensions range to few dozens has been studied widely in the visual analytic literature. However, very few studies focus on the analysis of high dimensional data [62]. Therefore, research efforts are required to design novel data reduction techniques, subspace selection and subspace projection techniques to tackle this limitation.

- *Usability*: Visualization toolkits designed for high dimensional data analysis are generally not very intuitive. These toolkits heavily involve user interaction to execute analysis tasks. Although design theories [63] [64] [65] and methodologies [66] [67] [68] [69] [70] [71] [72] have been developed to construct effective visualizations, research efforts are required to built usability evaluation methods to access usability of toolkits.

6 CONCLUSION

In high dimensional data, meaningful patterns are prominent only in sub-dimensional space. Therefore, in order to understand high dimensional data, it is very important to conceive subspace clustering results. Subspace clusters provide a divergent view on the same data. However, stems some problems like handling an enormous number of clusters and subspaces which can be possibly redundant and irrelevant. Visual analytics domain is proved to be an effective solution to interpret, analyze and navigate subspace clustering results. Many visualizations are designed that reveal interesting relationships in high dimensional data. This paper contributes by categorizing them and listing pros and cons of different method under each category. Summary of all the methods, discussed in this paper from different sections is provided in Table.2 which briefly discuss their core idea, visual aspect and applications or usage. Through this study, we deduced the best-suited visualizations for representing different aspects of comprehending high dimensional data. This is demonstrated in Table.3. This survey can help researchers from data mining and data visualization community to design advanced visual interactive systems to get an insight of high dimensional data .

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TABLE 2
Summary of all Methods

METHOD	CORE IDEA	VISUAL ASPECT			USAGE
SURFING	Finding all subspaces that are interesting for clustering and ranking using KNN distance. Subspace is interesting if hierarchical clustering structure is exhibited by data in that subspace.	-			(1) Identifies interesting subspaces in high dimensional data
Subspace Search to Make sense of Alternative Clustering	Construct candidate set of subspaces by leveraging dimensional and topological similarity in subspaces.	Scatterplots			(1) Removing redundancy in subspaces.
Finding Relevant Subspace for Clustering using Morphological Operators	Subspace is considered relevant if it has multiple Gaussian distribution spread over the attribute range.	Maxtree graph, scatterplot	representation, scatterplot	line PCA	(1) Identifies interesting subspaces in high dimensional data. (2) Reveal Quantitative information related to subspace like number of clusters, significance of clusters etc. (3) In depth exploration of ranked subspaces.
VISA	Compactly visualize subspace clusters in a comparable environment. It proposes a novel distance measure for estimating similarity between subspace clusters based on which MDS projection of subspace clusters is obtained.	MDS projection, Matrix visualization, mean and variance plot for dimensions of subspace cluster.			(1) Overview of subspace clusters. (2) Distance function which calculate (dis)similarity of subspace clusters. (3) Interestingness of subspace clusters.
ClustNails	Detailed inspection of subspace clusters	Spikes, HeatNails			(1) To examine cluster properties. (2) To draw cluster comparison.
S-ADViSED	Integrates well known visualizations to obtain overview of subspace clusters to assist epidemiologists.	Donut charts, MDS Image, Donut heatmaps, Mosaic charts, scatter plot, histogram, stacked error bars			(1) To examine cluster properties. (2) To draw cluster comparison.
SubVIS	To explore subspace clusters visually to gain insight	Heatmaps, MDS IMAGE, Bar-charts, aggregation tables, table lens			(1) To examine cluster properties. (2) To draw cluster comparison.
BEADS	Determine shape and size of higher dimensional clusters in 2-dimension using mathematical models.	Beads plot			(1) Determine shape of high dimensional cluster.
Heidi Matrix	Visualize proximity of every pair of point in dataset across various subspaces by building a matrix based on their knn relationship.	Heidi matrix			(1) Represents locality information between points across all subspaces. (2) Represents overlap of clusters in different subspaces. (3) Prominent set of subspaces can be determined.
Pattern trails	To order and arrange subspace projections side by side and then visualizing and interpreting pattern transition through linked representation. It aims to find structural changes in patterns across various subspace projections based on operations in attribute space.	Subspace projections			(1) Order the subspaces such that change in pattern can be effectively traced and interpreted across subspaces. (2) Analyze which attribute cause change in pattern.
Dimension Re-construction	To reconstruct dimensions using original subspace projections that preserve subspace cluster information.	multiple scatter plot projections			(1) Display and analyse subspace clusters structures clearly by interactive subspace construction. (2) Gives different view on same high dimensional data. (3) Preserve interesting cluster information. (4) Track clusters in different dimensions
SUBEVAL	Visually access subspace cluster quality	Inter-linked MDS projection visualization			(1) Defines criterias to visually access subspace quality. (2) Visualizes and compare quality of subspace clusters.

TABLE 3
Best suited visualizations for different aspects of comprehending high dimensional data

USAGE	VISUALIZATION	METHOD
Cluster representation	Donut Charts	S-ADViSED
Subspace projection ordering	Pattern trails Visualization	Pattern trails
Shape and size of clusters	Beads Visualization	BEADS
Relation between attributes/ Importance of attribute in subspace cluster	Spike View Visualization	ClustNails
Quality of subspace clustering results	Interlinked MDS Projection Visualization	SUBEVAL
Cluster Overlap	MDS projection + Heatmap Visualization	SubVIS
Dimensions Involved	Donut Charts	S-ADViSED
Ordering Techniques	HeatNail Visualization	ClustNails
Interestingness of subspace clusters	Matrix visualization	VISA
Distribution of data in dimension involved	Donut Charts	S-ADViSED
Filters to select subspace Clusters for analysis	-	SubVIS
Subspace Cluster Overview	MDS projection	VISA
Cluster object Overlap(In multiple sub-spaces)	Heidi Visualization	Heidi
Removing redundancy in subspace	-	Subspace Search to Make sense of Alternative Clustering
Variance in dimensions of subspace cluster	Donut Charts	S-ADViSED

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